Ensemble Learning

Ensemble Learning

- Machine learning model which combines the predictions of several machine learning models to improve the prediction accuracy of using a single classifier
- It is developed to overcome the limitations of using a single hypothesis machine learning model.

Principle

- To ensemble hypotheses of different models either by combining the collective predictions of different models into a single final model or by employing different models in a sequential fashion to increase the accuracy of learning process.
- It is used to solve both classification and regression problems by reducing bias and variance and to boost accuracy of models.

Applications

- Remote sensing
- Security
- Gene classification
- Image segmentation
- Video retrieval
- healthcare

Ensembling

- Process of combining many weak learners to produce a strong leaner that performs better than a single weak learner.
- To learn better with a set of models instead of making one model to learn better.

Ensembling methods

- Majority voting
  - Taking average or weighted average of multiple machine learning models or base learners
- Stacking
  - Uses predictions of multiple models as features to train a new model and then predict from that model
- To employ same machine learning model multiple times with different datasets and combine the hypothesis generated from these models

Bagging and Boosting

- Bagging
  - Method of ensembling simultaneously many independent models and averaging the predictions obtained from the different models.
  - It solves overfitting problem in single classifier
  - It minimizes variance
- Boosting
  - Employs different classifiers in a sequential fashion where the prediction made by a classifier is enhanced by boosting the weights for the data instances and consecutively the predicted output is given as input to next learning model.
  - It increases accuracy

Advantages

- Minimizes bias and variance of a weak learner.
- Reduces overfitting and underfitting of the model.
- Increases accuracy of prediction.
- Handles higher dimensionality data efficiently.

Disadvantages

- Does not give precise value for the classification and regression model.
- Error prone and sometimes leads to overfitting.
- Requires careful tuning of the number and types of features used.

Ensembling Techniques

- Ensemble models are generally classified as Parallel and Sequential ensemble models.
- Classified as ‘Homogenous’ and ‘Heterogenous’ ensemble models.
- In homogenous ensemble model, same type of base learners are used, whereas in Heterogenous ensemble model, different type of basic learners are used for ensembling.

Contd…

- In Parallel ensemble model, base learner learn simultaneously and independently.
- Include Voting, Bootstrap, Resampling, Bagging and Random forest.
- In sequential ensemble model, they learn sequentially.
- AdaBoost, Gradient Tree Boosting and XGBoost also called as boosting model.
- Parallel ensemble learners are time and space efficient compared to sequential ensemble learners.

Voting and Bootstrap Resampling

- **Voting** is the process of combining multiple ML models or using same model multiple times as base learners.
- In regression problem it takes average of their predictions, whereas in classification problem the maximum vote of their prediction.
- **Bootstrap** is the process of creating many bootstrap samples using a method called Random sampling from training dataset.
- Each sample contains 63.2% distinct tuples and remaining are duplicate tuples from the original dataset.

Bagging

- Also called as Bootstrap Aggregation.
- It combines both bootstrapping and aggregation.
- Trains M independent weak classifier with each classifier containing bootstrap samples
- If the target is categorical variable - majority voting continuous variable - averaging

Random Forest

- Instead of training different model with same features , this model chooses random set of features to construct decision tree.
- Collection of trees with each tree constructed with random samples, random set of features and random split features.
- Learning model is decision tree with subset of features, ensemble model improves the efficiency with quick searching.

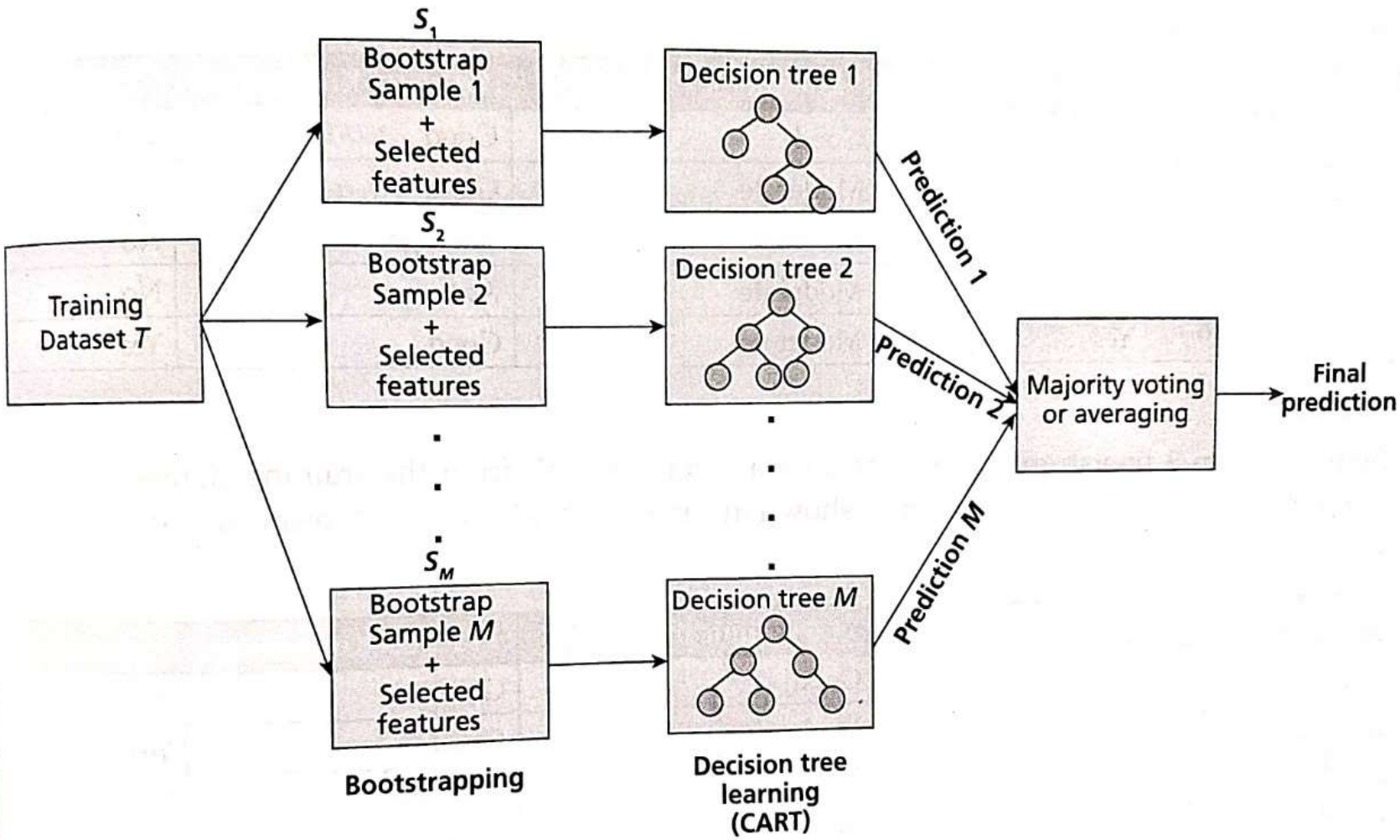

Figure 12.1: Random Forest Model

Random Forest

- Step 1: Select random samples from a given data or training set.
- Step 2: This algorithm will construct a decision tree for every training data.
- Step 3: Voting will take place by averaging the decision tree.
- Step 4: Finally, select the most voted prediction result as the final prediction result.

RF Algorithm Steps

- Step 1: Create M bootstrap samples of N data instances using Random Sampling with replacement
- Step 2: Choose number of decision trees to be constructed DT_M
- Step 3: Choose the number of features to be used in each tree. Normally sqrt(D) randomly chosen features are taken
- Step 4: Choose the depth of the decision tree. All trees are to be same depth.
- Step 5: Repeat M times. For each tree
  - Choose a bootstrap sample S_i of N data instances
  - Compute the best split feature to be placed at each node
  - Construct decision tree using CART. Do not prune the tree
- Step 6: Combine predictions from all classifiers by taking average for regression problem and majority voting for classification problem.

Advantages and Disadvantages

- Can solve both classification and regression problems
- Reduces overfitting and variance compared to single decision tree
- Provides more accuracy
- Performs well even with smaller dataset
- Disadvantages
  - Increased complexity
  - Uses more computational resources

Incremental Ensemble Models

- Use the predictions that are learnt from the previous learner as features to train a meta learner and make final predictions
- Types
  - Stacking
  - Cascading

Stacking

- Meta modelling technique
- Includes two types of learners called base-learners and meta-learner
- Process of ensembling multiple heterogeneous machine learning models or base learners to make predictions and then use these predictions as features to train the second level meta learner model
- Base learners use normal ML models whereas meta learner fits on predictions of the base learner.
- Meta leaner can be meta-regressor or meta-classifier
- Further extension of stacking is called multi-level stacking

Cascading

- Mulistage learning technique with a sequence of classifiers ordered in terms of increasing complexity and specificity such that early classifiers are simple and general whereas later ones are more complex and specific.
- It moves to the next stage of learning only if the preceding learner's prediction is not confident.
- Complexity increases as more models are added to the cascade.

Sequential Ensemble Models

- Use multiple weak learners sequentially and the error exhibited by a weak learner is used to give importance to data instances which are further retrained by the next weak learner.

AdaBoost

- Also called Adaptive Boosting is a popular boosting algorithm and it is non-linear classifier.
- Sequential ensemble model that trains multiple weak classifiers sequentially
- It would train one weak classifier at a time, so that the errors made by poorly performing weak classifiers are reduced when they are trained sequentially

- AdaBoost is a greedy algorithm optimizing weights of data instances in the training dataset by adding a weak classifier at each step.
- Each weak classifier is trained with a random bootstrap sample drawn from the training dataset.
- Initially weights are assigned to all data instances in the training dataset with equal probability.
- If there are 10 data instances in training dataset, the probability of choosing a data instance will be 1/10.
- After training by one weak classifier, the weights of all data instances are boosted based on the accuracy of classification done by it.
- The idea of boosting is to minimize the errors made by a weak classifier by increasing the weights of misclassified data instances and decreasing the weights of correctly classified data instances.
- The updated probabilities of data instances are reflected in the training dataset and these instances get trained with the next classifier.

- Each weak classifier is also assigned a weight after training, based on the accuracy of the classification.
- If the classifier is more accurate, it is given more weight
- A 50% accurate classifier is given a weight of 0.
- If the classifier has less than 50% accuracy, it gets a negative weight.

- There are many weak classifiers such as decision trees, decision stumps, multilayer perceptron.
- Linear classifiers such as decision stumps are on-level decision trees which perform poorly with low accuracy.
- The depth of decision stumps is 1 and it has two leaves.
- The final classifier makes a linear combination of all the predictions made by the weak classifiers.

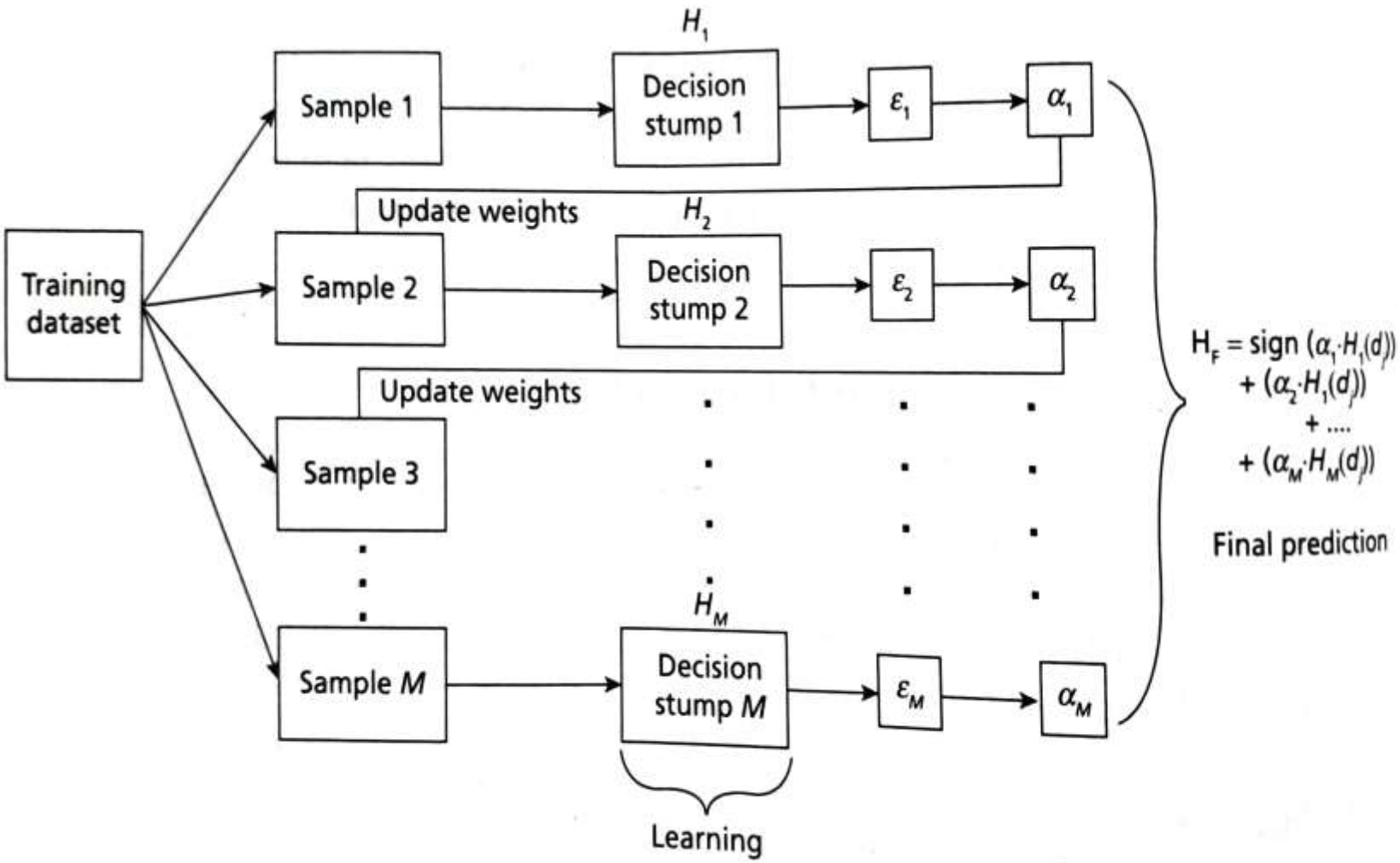

Figure 12.5: Working of AdaBoost Algorithm

AdaBoost Algorithm Steps

- Step1: Assign uniform weights 1/N to all data instances in the training dataset
- Step2: Repeat for each weak classifier
  - Train a weak classifier Hi with random bootstrap samples from training dataset T
  - Compute the weighted error ε_i of H_i on current training dataset
  - Compute the weight of each weak classifier α_i based on error rate.
  - Calculate the normalizing factor Z_i
  - Update the weight of all data instances
- Step 3: Compute final predicted value for each data instance. It is the linear combination of weighted average predictions of all the classifiers multiplied by weight α_i computed for each classifier